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Report: **AM2024-0074**

# **BET N<sub>2</sub> sorption measurements on particle samples of natural origin**

## **Short Summary 7**

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Client:

Curia Germany GmbH  
Industriepark Höchst D569  
65926 Frankfurt am Main

Contractor:

Fraunhofer-Institut für Silicatforschung ISC  
Zentrum für Angewandte Analytik ZAA  
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Project Management:

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# Documentation Overview

## Contact

on the part of the client

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on the part of the contractor

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## ISC Offer Number

2210-2024-51000-45-02801-00010 and  
2210-2024-51000-45-02801-00019

## Order of the Client

signature on offer 2210-2024-51000-45-02801-00010  
dated 26 July 2024 and  
signature on offer 2210-2024-51000-45-02801-00019  
dated 09 September 2024

# Place of Performance and Processing Time

## Place of Performance

Center for Applied Analytics  
Fraunhofer Institute for Silicate Research  
Neunerplatz 2  
97082 Würzburg  
Germany

## Processing Time

|                       |            |
|-----------------------|------------|
| Sample receipt        | 09.12.2024 |
| Begin of the analysis | 16.12.2024 |
| End of the analysis   | 18.12.2024 |
| Date of report        | 19.12.2024 |

# Sample Overview

| Sample No. | Description                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------------|
| P07        | Activated Charcoal Pyrolysate – Batch: MB3001A –<br>Date: 04.12.2024                                                           |
| P08        | Activated Charcoal Pyrolysate – Batch: MB3004A –<br>Date: 04.12.2024                                                           |
| P09        | Activated Charcoal Pyrolysate – Batch: MB3001A/3004A –<br>Date: 04.12.2024                                                     |
| P10        | Source: MKCV4555 – Single-layer graphene sheets for battery –<br>Bio-sourced (Sigma Aldrich / Merck)                           |
| P11        | Source: MKCV7546 – Graphene powder, electrical conductivity $> 10^3$ S/m –<br>avg. no. of layers $< 3$ (Sigma Aldrich / Merck) |

# Procedure of Testing

- The powder samples were used as received for the BET N<sub>2</sub> sorption measurements.
- To determine the surface area, the powders were measured in a Quantachrome Autosorb-3B device. The amount of the analysed material was quite little, so the results should be interpreted with a grain of salt.

# BET Surface Area

| Sample No. | Mass<br>[g] | BET Surface Area<br>[m <sup>2</sup> /g] | Langmuir Surface Area<br>[m <sup>2</sup> /g] |
|------------|-------------|-----------------------------------------|----------------------------------------------|
| P07        | 0.1086      | 1.613 x 10 <sup>3</sup>                 | 7.763 x 10 <sup>3</sup>                      |
| P08        | 0.0767      | 1.491 x 10 <sup>3</sup>                 | 2.032 x 10 <sup>3</sup>                      |
| P09        | 0.1384      | 1.471 x 10 <sup>3</sup>                 | 1.983 x 10 <sup>3</sup>                      |
| P10        | 0.2208      | 1.112 x 10 <sup>3</sup>                 | 1.421 x 10 <sup>3</sup>                      |
| P11        | 0.2477      | 1.942 x 10 <sup>1</sup>                 | 2.442 x 10 <sup>1</sup>                      |

# Final Remarks

- The analyses are carried out by specifically qualified employees and by the use of up-to-date analysis machines. Nevertheless, we are not able to provide guarantee for the correctness of the drawn conclusion.
- The results only refer to the provided and examined samples.
- Sample taking and shipping of the samples to the ZAA took place under the responsibility of the client.
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End of report AM2024-0074 – Short Summary 7

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